

Raju Lama

Assignment 6: Generate Packets to Scan Targets

Objective

The goal of this assignment is to create ICMP, TCP, and UDP packets and see them in Wireshark.

Tools Used

Wireshark

Terminal on Mac

ping command

curl command

nc command

ICMP Packets

Command used:

ping 8.8.8.8

This command sends ICMP packets.

In Wireshark, I saw echo request and echo reply packets.

This shows ICMP is working.

TCP Packets

Command used:

curl http://scanme.nmap.org

This command sends TCP packets.

In Wireshark, I saw TCP packets like SYN and ACK.

I also saw a web request and response.

UDP Packets

Command used:

nc -u scanme.nmap.org 80

I typed hello and pressed enter.

In Wireshark, I saw UDP packets sent from my computer.

There was no reply, which is normal for UDP.

Problem Faced

Raju Lama

I tried to use Nping but it did not work on my Mac. So I used other commands to create the same types of packets.

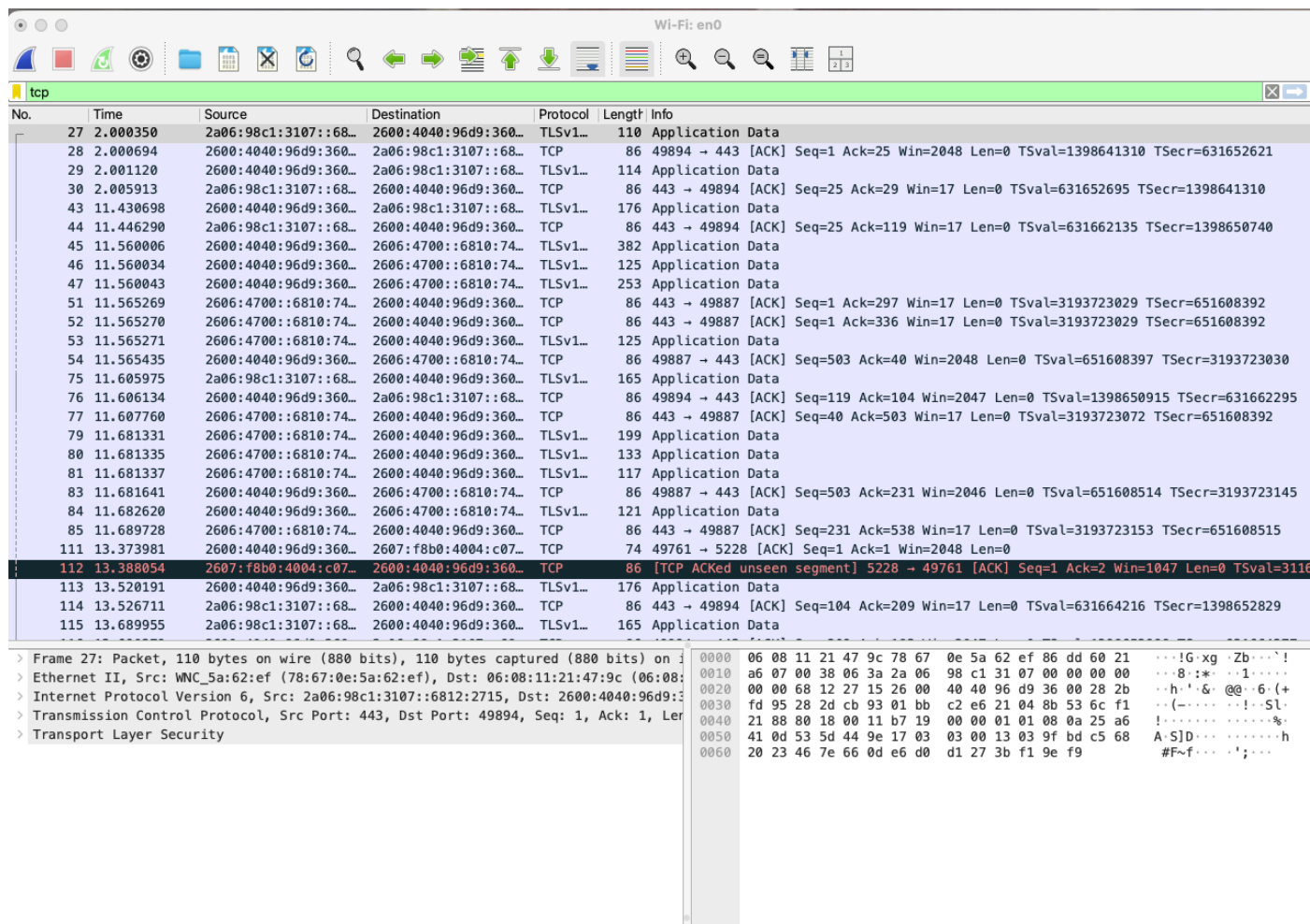
Conclusion

I was able to create and capture ICMP, TCP, and UDP packets. Wireshark showed all the packets clearly.

The packets captured in UDP Wireshark screenshot show my own IP address, which is different from other students.

Screenshots

ICMP Wireshark



ICMP Terminal

Raju Lama

● ● ● rajulama — -zsh — 125x33

```
Last login: Sun Apr 19 09:36:14 on ttys000
/Users/rajulama/.zprofile:export:1: not valid in this context: PATH,
rajulama@MacBookPro ~ % nping --icmp-type 8 8.8.8.8
```

```
Starting Nping 0.7.98 ( https://nmap.org/nping ) at 2026-04-19 09:41 EDT
libnsock mksock_bind_addr(): Bind to 0.0.0.0:0 failed (IOD #1): Invalid argument (22)
SENT (0.0026s) Starting TCP Handshake > 8.8.8.8:80
libnsock mksock_bind_addr(): Bind to 0.0.0.0:0 failed (IOD #2): Invalid argument (22)
SENT (1.0061s) Starting TCP Handshake > 8.8.8.8:80
libnsock mksock_bind_addr(): Bind to 0.0.0.0:0 failed (IOD #3): Invalid argument (22)
SENT (2.0092s) Starting TCP Handshake > 8.8.8.8:80
libnsock mksock_bind_addr(): Bind to 0.0.0.0:0 failed (IOD #4): Invalid argument (22)
SENT (3.0115s) Starting TCP Handshake > 8.8.8.8:80
libnsock mksock_bind_addr(): Bind to 0.0.0.0:0 failed (IOD #5): Invalid argument (22)
SENT (4.0140s) Starting TCP Handshake > 8.8.8.8:80
```

```
Max rtt: N/A | Min rtt: N/A | Avg rtt: N/A
TCP connection attempts: 5 | Successful connections: 0 | Failed: 5 (100.00%)
Nping done: 1 IP address pinged in 5.02 seconds
rajulama@MacBookPro ~ % sudo nping --icmp-type 8 8.8.8.8
Password:
```

```
Starting Nping 0.7.98 ( https://nmap.org/nping ) at 2026-04-19 09:47 EDT
Assertion failed: (e != NULL), function netutil_eth_send, file netutil.cc, line 1039.
zsh: abort      sudo nping --icmp-type 8 8.8.8.8
rajulama@MacBookPro ~ % sudo nping --tcp -p 80 scanme.nmap.org
```

```
Starting Nping 0.7.98 ( https://nmap.org/nping ) at 2026-04-19 09:47 EDT
Assertion failed: (e != NULL), function netutil_eth_send, file netutil.cc, line 1039.
zsh: abort      sudo nping --tcp -p 80 scanme.nmap.org
rajulama@MacBookPro ~ %
```

TCP Wireshark

Wireshark packet capture showing a TCP connection to port 80. The packet list shows a SYN packet (482) and a subsequent ACK packet (492). The packet details for the ACK packet show the sequence number 4920 and the acknowledgment number 80. The packet bytes show the raw data of the ACK packet.

No.	Time	Source	Destination	Protocol	Length	Info
482	27.681286	2600:4040:96d9:360...	2600:3c01::f03c:91...	TCP	98	49920 → 80 [SYN, ECE, CW] Seq=0 Win=65535 Len=0 MSS=1440 WS=64 TSval=943105310 TSecr=0 ...
483	27.749033	2600:3c01::f03c:91...	2600:4040:96d9:360...	TCP	94	80 → 49920 [SYN, ACK, ECE] Seq=0 Ack=1 Win=64260 Len=0 MSS=1440 SACK_PERM TSval=18920226...
484	27.749443	2600:4040:96d9:360...	2600:3c01::f03c:91...	TCP	86	49920 → 80 [ACK] Seq=1 Ack=1 Win=131392 Len=0 TSval=943105378 TSecr=1892022698
485	27.749604	2600:4040:96d9:360...	2600:3c01::f03c:91...	HTTP	164	GET / HTTP/1.1
486	27.828504	2600:3c01::f03c:91...	2600:4040:96d9:360...	TCP	86	80 → 49920 [ACK] Seq=1 Ack=79 Win=64256 Len=0 TSval=1892022767 TSecr=943105378
487	27.828509	2600:3c01::f03c:91...	2600:4040:96d9:360...	TCP	1514	80 → 49920 [ACK] Seq=1 Ack=79 Win=64256 Len=1428 TSval=1892022768 TSecr=943105378 [TCP P...
488	27.828516	2600:3c01::f03c:91...	2600:4040:96d9:360...	TCP	1514	80 → 49920 [PSH, ACK] Seq=1429 Ack=79 Win=64256 Len=1428 TSval=1892022768 TSecr=94310537...
489	27.828520	2600:3c01::f03c:91...	2600:4040:96d9:360...	TCP	1514	80 → 49920 [ACK] Seq=2857 Ack=79 Win=64256 Len=1428 TSval=1892022768 TSecr=943105378 [TC...
490	27.828523	2600:3c01::f03c:91...	2600:4040:96d9:360...	TCP	1514	80 → 49920 [PSH, ACK] Seq=4285 Ack=79 Win=64256 Len=1428 TSval=1892022768 TSecr=94310537...
491	27.828527	2600:3c01::f03c:91...	2600:4040:96d9:360...	TCP	1514	80 → 49920 [ACK] Seq=5713 Ack=79 Win=64256 Len=1428 TSval=1892022775 TSecr=943105378 [TC...
492	27.828531	2600:3c01::f03c:91...	2600:4040:96d9:360...	HTTP	142	HTTP/1.1 200 OK (text/html)
493	27.828903	2600:4040:96d9:360...	2600:3c01::f03c:91...	TCP	86	49920 → 80 [ACK] Seq=79 Ack=7197 Win=124224 Len=0 TSval=943105457 TSecr=1892022775
494	27.829188	2600:4040:96d9:360...	2600:3c01::f03c:91...	TCP	86	[TCP Window Update] 49920 → 80 [ACK] Seq=79 Ack=7197 Win=131072 Len=0 TSval=943105458 TS...
495	27.830929	2600:4040:96d9:360...	2600:3c01::f03c:91...	TCP	86	49920 → 80 [FIN, ACK] Seq=79 Ack=7197 Win=131072 Len=0 TSval=943105460 TSecr=1892022775
496	27.897408	2600:3c01::f03c:91...	2600:4040:96d9:360...	TCP	86	80 → 49920 [FIN, ACK] Seq=7197 Ack=80 Win=64256 Len=0 TSval=1892022847 TSecr=943105460
497	27.897790	2600:4040:96d9:360...	2600:3c01::f03c:91...	TCP	86	49920 → 80 [ACK] Seq=80 Ack=7198 Win=131072 Len=0 TSval=943105526 TSecr=1892022847

> Frame 482: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface 0
> Ethernet II, Src: 06:08:11:21:47:9c (06:08:11:21:47:9c), Dst: WNC_5a:62:ef (78:67:2d:5a:62:ef), Protocol: TCP
> Internet Protocol Version 6, Src: 2600:4040:96d9:3600:282b:fd95:282d:cb93, Dst: 2600:3c01::f03c:9100:0000:0000
> Transmission Control Protocol, Src Port: 49920, Dst Port: 80, Seq: 0, Len: 0

0000 78 67 0e 5a 62 ef 06 08 11 21 47 9c 86 dd 60 0a xg Zb... !G...
0010 06 00 00 2c 06 40 26 00 40 00 96 d9 36 00 28 2b ... ,@& @ @ 6 (+
0020 fd 95 28 2d cb 93 26 00 3c 01 00 00 00 00 f0 3c ... (-& <<
0030 91 ff fe 18 bb 2f c3 00 00 50 3e 36 c8 39 00 00 ... /... P>6-9..
0040 00 00 b0 c2 ff ff a5 18 00 00 02 04 05 a0 01 03P>6-9..
0050 03 06 01 01 08 0a 38 36 a5 1e 00 00 00 00 04 0286
0060 00 00

TCP Terminal

● ● ● ● rajulama — zsh — 125x87

```
rajulama@MacBookPro ~ % curl http://scanme.nmap.org
<!DOCTYPE html>
<html lang="en">
<head>
<title>Go ahead and ScanMe!</title>
<meta name="viewport" content="width=device-width,initial-scale=1">
<meta name="theme-color" content="#2A0D45">
<link rel="preload" as="image" href="/images/sitelogo.png" imagesizes="168px" imagesrcset="/images/sitelogo.png, /images/site
logo-2x.png 2x">
<link rel="preload" as="image" href="/shared/images/nst-icons.svg">
<link rel="stylesheet" href="/shared/css/nst.css?v=2">
<script async src="/shared/js/nst.js?v=2"></script>
<link rel="stylesheet" href="/shared/css/nst-foot.css?v=2" media="print" onload="this.media='all'">
<link rel="stylesheet" href="/site.css">
<!--Google Analytics Code-->
<link rel="preload" href="https://www.google-analytics.com/analytics.js" as="script">
<script>
(function(i,s,o,g,r,a,m){i['GoogleAnalyticsObject']=r;i[r]=i[r]||function(){
(i[r].q=i[r].q||[]).push(arguments)},i[r].l=1*new Date();a=s.createElement(o),
m=s.getElementsByName(o)[0];a.async=1;a.src=g;m.parentNode.insertBefore(a,m)
})(window,document,'script','https://www.google-analytics.com/analytics.js','ga');
ga('create','UA-11009417-1','auto');
ga('send','pageview');
</script>
<!--END Google Analytics Code-->
<META NAME="ROBOTS" CONTENT="NOARCHIVE">
<link rel="shortcut icon" href="/shared/images/tiny-eyeicon.png" type="image/png">
</head>
<body><div id="nst-wrapper">

<div id="menu">
<div class="blur">
<header id="nst-head">

<a id="menu-open" href="#menu" aria-label="Open menu">

</a>
<a id="menu-close" href="#" aria-label="Close menu">

</a>

<a id="nst-logo" href="/" aria-label="Home page">
</a>

<nav id="nst-gnav">
<a class="nlink" href="https://nmap.org/">Nmap.org</a>
<a class="nlink" href="https://npcap.com/">Npcap.com</a>
<a class="nlink" href="https://seclists.org/">Seclists.org</a>
<a class="nlink" href="https://sectools.org/">Sectools.org</a>
<a class="nlink" href="https://insecure.org/">Insecure.org</a>
</nav>

<form class="nst-search" id="nst-head-search" action="/search/">
```

UDP Wireshark

Capturing from Wi-Fi: en0

No.	Time	Source	Destination	Protocol	Length	Info
2	0.183315	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
3	0.183381	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
4	0.183383	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
5	0.183386	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
6	0.183390	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
7	0.183394	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
8	0.183397	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
9	0.183414	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
10	0.183417	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
11	0.183421	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
12	0.183431	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
13	0.183434	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
14	0.192128	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
15	0.192169	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
16	0.194340	2a06:98c1:3100::68...	2600:4040:96d9:360...	UDP	86	443 → 49217 Len=24
17	0.198848	2a06:98c1:3100::68...	2600:4040:96d9:360...	UDP	86	443 → 49217 Len=24
18	0.198851	2a06:98c1:3100::68...	2600:4040:96d9:360...	UDP	86	443 → 49217 Len=24
19	0.199223	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
20	0.199347	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	465	49217 → 443 Len=403
21	0.208156	2a06:98c1:3100::68...	2600:4040:96d9:360...	UDP	87	443 → 49217 Len=25
22	0.351539	2a06:98c1:3100::68...	2600:4040:96d9:360...	UDP	774	443 → 49217 Len=712
23	0.353648	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	107	49217 → 443 Len=45
28	2.752068	192.168.1.1	192.168.1.185	MDNS	86	Standard query 0x1be9 PTR 185.1.168.192.in-addr.arpa, "QM" question
49	2.761986	192.168.1.185	192.168.1.1	MDNS	123	Standard query response 0x1be9 PTR 185.1.168.192.in-addr.arpa, "QM" question PTR Rajus
56	8.189260	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
57	8.189323	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230
58	8.189327	2600:4040:96d9:360...	2a06:98c1:3100::68...	UDP	1292	49217 → 443 Len=1230

> Frame 28: Packet, 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface
 > Ethernet II, Src: WNC_5a:62:ef (78:67:0e:5a:62:ef), Dst: 06:08:11:21:47:9c (06:08:11:21:47:9c)
 > Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.185
 > User Datagram Protocol, Src Port: 40001, Dst Port: 5353
 > Multicast Domain Name System (query)

```

0000  06 08 11 21 47 9c 78 67 0e 5a 62 ef 08 00 45 00  ...!G.xg.Zb...E.
0010  00 48 c5 f6 40 00 40 11 f0 a3 c0 a8 01 01 c0 a8  .H:@@:.....
0020  01 b9 9c 41 14 e9 00 34 e4 b6 1b e9 04 00 00 01  .A~4.....
0030  00 00 00 00 00 00 03 31 38 35 01 31 03 31 36 38  ....185.1.168
0040  03 31 39 32 07 69 6e 2d 61 64 64 72 04 61 72 70  .192.in-addr.arp
0050  61 00 00 0c 00 01                                     a.....

```

UDP Terminal

```

rajulama --zsh-- 125x23

<a class="nlink" href="https://twitter.com/nmap" title="Visit us on Twitter">

</a>
<a class="nlink" href="https://facebook.com/nmap" title="Visit us on Facebook">

</a>
<a class="nlink" href="https://github.com/nmap/" title="Visit us on Github">

</a>
<a class="nlink" href="https://reddit.com/r/nmap/" title="Discuss Nmap on Reddit">

</a>
</div>
</div>
</footer>

</div><!-- wrapper -->
</body>
</html>

[rajulama@MacBookPro ~ % nc -u scanme.nmap.org 80
hello
rajulama@MacBookPro ~ % ]

```